

Curriculum Vitæ – Alberto De Rosa

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CURRENT POSITION

Postdoctoral Fellowship, University of Alberta, Edmonton, Alberta, Canada. February 2023 to December 2026.

Supervisors: Matina Kalcounis-Rueppell and Erin Bayne.

Main Topics: student mentoring, bioacoustics, conservation biology, passive acoustic monitoring, small mammal research, habitat suitability, ultrasonic bat acoustic data, landscape ecology, community-led research, teaching, and quantitative ecology.

Additional technical role: fieldwork — support for reproducible computational workflows and management of OpenStack virtual resources in the current lab.

EDUCATION

Doctorate in Zoology, Massey University, Te Kunenga ki Pūrehuroa, Palmerston North, Aotearoa New Zealand. August 2022.

Supervisors: Isabel Castro, Stephen Marsland, Weihong Ji.

Dissertation: “On Kiwi (*Apteryx mantelli*, Bartlett) vocal behaviour and activity: relations to population densities, and applications to conservation”.

Main Topics: statistical ecology, bioacoustics, conservation biology, passive acoustic monitoring, behavioural ecology, paired sampling, playback technique.

Master of Science in Biodiversity and Evolution, University of Bologna, Bologna, Italy. March 2014.

Final score: 110/110, cum laude.

Supervisor: Maria Vallisneri.

Dissertation: “Inter-specific foraging behaviour analysis in a mixed group of birds, in a semi-natural area in the Wild Birds Recovery Centre of Faenza, Italy”.

Main Topics: avian population ecology, foraging ecology, biodiversity monitoring, conservation acts, population genetics of Mediterranean corals *Balanophyllia europaea*.

Bachelor of Science in Biological Sciences, University of Bologna, Bologna, Italy. October 2007.

Final score: 91/110.

Supervisor: Francesco Zaccanti.

Dissertation: “The wolves in the Monti Sibillini National Park: a snowtracking survey”.

Main Topics: wolf *Canis lupus* population ecology, snowtracking, anthropic environmental impact, movement ecology, and biogeography.

High School Diploma, Technical High School in Information Technology, ITIS “Montani”, Fermo, Italy. June 2004.

Final score: 84/100.

Final project: high- and low-level software development to control a step-by-step motor-driven mechanical arm over the internet with webcam integration.

RESEARCH INTERESTS

Conservation biology, statistical ecology, citizen science with special interest in collaboration and work with local community groups, passive acoustic monitoring, bioacoustics, evolution, speciation, and adaptation.

RESEARCH OUTPUTS

ARTICLES PUBLISHED

De Rosa, A., Castro, I., and Marsland, S. 2021. The playback technique in avian fieldwork contexts: a systematic review and recommendations for best practice. *Ibis*. doi:10.1111/ibi.13033.

Castro, I., De Rosa, A., Priyadarshani, N., Bradbury, L., and Marsland, S. 2019. Experimental test of birdcall detection by autonomous recorder units and by human observers using broadcast. *Ecology and Evolution* 9:2376–2397. doi:10.1002/ece3.4775.

CONFERENCE AND ENGAGEMENT OUTPUTS

- Conklin Métis Community, Fortna, P., De Rosa, A., Saini, A., Meyhoff, S., Evans, D., and Bayne, E. Industrial noise and moose call transmission in northeastern Alberta: pilot synthesis and continuation guide. Condition of the Territory: Community-Based Monitoring in the Oil Sands gathering. Virtual. March 2026.
- De Rosa, A., Hurtado, C., Bayne, E., and Kalcounis-Rueppell, M. Modelling bat echolocation data at provincial scales: structure, signal uncertainty, and spatial grain. North American Society of Bat Research Conference. Edmonton, Alberta, Canada. October 2025.
- De Rosa, A., Hurtado, C., Bayne, E., and Kalcounis-Rueppell, M. Model structure selection to minimise uncertainty in modelling hoary bat echolocation detections in Alberta. The Wildlife Society. Edmonton, Alberta, Canada. October 2025.
- De Rosa, A., Castro, I., and Marsland, S. Implementing spatial mark-resight models with acoustic activity data described through environmental covariates. International Bioacoustic Congress. Sapporo, Hokkaido, Japan. November 2023.
- De Rosa, A., Hurtado, C., Kalcounis-Rueppell, M., and Bayne, E. Landscape-level correlates of bats in the boreal region of Alberta, Canada. North American Society of Bat Research Conference. Winnipeg, Manitoba, Canada. October 2023.
- Marsland, S., De Rosa, A., and Castro, I. Animal-borne acoustic recorders can inform consistent abundance estimates based on passive acoustic monitoring. New Zealand Ecological Society Conference. Dunedin, Otago, Aotearoa New Zealand. November 2022.
- Castro, I., Feenstra, E., De Rosa, A., and Marsland, S. Smile! You are on camera: use of trail cameras to estimate the abundance of a cryptic, unmarked species, Stewart Island kiwi/Rakiura tokoeka. New Zealand Ecological Society Conference. Dunedin, Otago, Aotearoa New Zealand. November 2022.
- De Rosa, A., Olsen, D., Castro, I., and Marsland, S. Pairing community and individual vocal activity data can inform consistent abundance estimates using animal-borne acoustic recorders. North American Congress for Conservation Biology. Reno, Nevada, United States of America. July 2022.
- De Rosa, A., Olsen, D., Castro, I., and Marsland, S. Animal-borne acoustic recorders inform mark-resight models for abundance estimates in North Island brown kiwi, *Apteryx mantelli*. International Statistical Ecology Congress. Cape Town, South Africa. June 2022.
- De Rosa, A., Olsen, D., Castro, I., and Marsland, S. Using animal-borne acoustic recorders to inform abundance estimates in kiwi. National Kiwi Hui. Lyttelton, Canterbury, Aotearoa New Zealand. June 2022.
- De Rosa, A., Castro, I., Olsen, D., and Marsland, S. Miniature animal-borne acoustic recording units as aids to investigate vocal behaviour and derive population density estimations. Australasian Ornithological Conference. Auckland, Aotearoa New Zealand. February 2022.
- De Rosa, A., Olsen, D., Castro, I., and Marsland, S. Miniature acoustic recorders embedded into kiwi transmitters: challenges and opportunities for behavioural ecology and conservation. Ecoacoustic Congress. Urbino, Italy. June 2021.
- De Rosa, A., White, A., Castro, I., and Marsland, S. Hey, R. U. There? Autonomous recording units as means to monitor sounds in a spatially explicit fashion. Statistical Ecology and Environmental Management Conference. Wellington, Aotearoa New Zealand. December 2019.
- Marsland, S., Priyadarshani, N., Juodakis, J., Castro, I., De Rosa, A., Listanti, V., and Feenstra, E. AviaNZ: software and research into acoustic monitoring of birds. Birds New Zealand Conference. Wellington, Aotearoa New Zealand. June 2019.
- Castro, I., Marsland, S., Priyadarshani, N., Juodakis, J., De Rosa, A., Listanti, V., and Feenstra, E. Unobtrusive monitoring of kiwi. Birds New Zealand Conference. Wellington, Aotearoa New Zealand. June 2019.

TEACHING EXPERIENCE AND QUALIFICATIONS

Topics: teaching and AI, assessments, student and community engagement.

Course: Festival of Teaching.

Date: April 2026.

Institution: University of Alberta.

Address: Lister Conference Centre, Edmonton, Alberta, Canada T6G 2X2.

Website: ualberta.ca.

Skills: scholarship of teaching, course alignment, instructional materials, learning outcomes, and assessment.
Course: Foundations of Teaching.
Date: August 2021 to October 2021.
Taught by: A. Graeme Pate.
Institution: University of Alberta.
Address: 5-02 Cameron Library, Edmonton, Alberta, Canada T6G 2J8.
Website: ualberta.ca.

Skills: andragogy, effective teaching, and reflective writing.
Course: Advance Higher Education, Introduction to University Teaching Course, Associate Fellow.
Date: July 2021 to October 2021.
Accreditation awarded: 3 December 2021.
Taught by: Linda Rowan and Fiona Murray.
Institution: Massey University.
Address: Riddet Road, Palmerston North 4410, Manawatū-Whanganui, Aotearoa New Zealand.
Website: massey.ac.nz.

From: 1 August 2021. **To:** 13 October 2021.
Employer: Massey University.
Address: Riddet Road, Palmerston North 4410, Manawatū-Whanganui, Aotearoa New Zealand.
Website: massey.ac.nz.
Position: Guest Speaker.
Duties: Teaching and class preparation.
Skills: engaging in interactive learning, evaluating student performance, and student-centred learning.

From: 1 August 2019. **To:** 30 November 2019.
Employer: Massey University.
Address: Riddet Road, Palmerston North 4410, Manawatū-Whanganui, Aotearoa New Zealand.
Website: massey.ac.nz.
Position: Lecturer.
Duties: Teaching and class preparation.
Skills: management of online class materials and interaction with tertiary students.

From: 3 October 2016. **To:** 30 June 2017.
Employer: Gruppo Umana Solidarietà, Macerata, Italy.
Address: Via della Pace 5, 62100 Macerata, Italy.
Website: gusitalia.it.
Position: Italian teacher.
Duties: Class management, preparation of individual study plans, teaching, and social introduction.
Skills: management of sensitive social situations, class and individual study plan design, delivery, and management.

Course: Advanced Models in Ecology.
Institution: University of Alberta.
Date: January to April 2024.
Activity: Co-designed, co-developed, and co-delivered this non-credit graduate course with Lionel Leston and Andrew Crosby.

FUNDING

Date: April 2021.
Funder: Ornithological Society of New Zealand.
Website: birdsnz.org.nz/funding/david-medway-scholarship/dms-projects.
Project: Kiwi (*Apteryx mantelli*) vocal behaviour and activity: population densities, and applications to conservation.
Amount: NZ\$5,000.

Date: 14 July 2020.
Funder: Kiwis for Kiwi.
Website: kiwisforkiwi.org.
Project: Translating numbers of calls into numbers of birds using cutting edge technology to improve Kiwi Call Counts.
Amount: NZ\$15,895.

Date: 26 January 2018.
Funder: Te Pūnaha Matatini.
Website: tepunahamatatini.ac.nz.

Project/role: PhD Scholarship.
Amount: NZ\$81,900 plus university fees.
Date: 26 July 2014.
Funder: Utah State University.
Website: usu.edu.
Project/role: Research assistantship, 12 months.
Amount: US\$24,000.

PROFESSIONAL ORGANISATIONS

International Bioacoustic Society. Since 14 April 2023.
North American Society for Bat Research. Since 1 May 2023.
The Wildlife Society. Since 2 July 2025.

OTHER RELEVANT JOB EXPERIENCES AND SKILLS

Course: Bat Acoustic Identification Workshops.
From: 27 January 2025. **To:** 31 January 2025.
Taught by: Vesper Bat Detection Services.
Skills: bat acoustic identification.

Course: H2S Awareness.
From: 5 August 2023.
Taught by: Canada Safety Training Centre.

Course: Defensive and Four Wheel Driving Accreditation.
From: 19 June 2023. **To:** 21 June 2023.
Taught by: Kenji Kinoshita.
Address: Human Resources, Health, Safety + Environment, University of Alberta.

Course: Bear Awareness and Safety.
From: 16 May 2023.
Taught by: Erin Bayne.
Address: Field Research Office, 3-36 Tory Building, University of Alberta.

Course: Wilderness First Aid.
From: 26 April 2023. **To:** 28 April 2023.
Taught by: Kris Porlier.
Address: Field Research Office, 3-36 Tory Building, University of Alberta.

Course: ATV Driving.
From: April 2023.
Taught by: Kenji Kinoshita.
Address: Human Resources, Health, Safety + Environment, University of Alberta.

Course: Intermediate First Aid with CPR C and AED.
From: 1 April 2023. **To:** 2 April 2023.
Taught by: James Garnet.
Address: St. Mark James Training, 7915 104 St NW #200, Edmonton, Alberta T6E 4E1.

Skills: use of trail cameras to evaluate biodiversity in agricultural settings.
From: May 2022.
Employer: Plant & Food Research.
Address: Batchelar Road, Fitzherbert, Palmerston North 4474, Aotearoa New Zealand.
Position: Consultant.
Duties: Experimental design, data extraction, and delivery of results.

Skills: advanced project management, deadline management, and Rcpp programming.
From: May 2022.
Employer: Victoria University of Wellington.
Address: Kelburn, Wellington, Aotearoa New Zealand.
Position: Research Assistant.
Duties: Translation of R code to Rcpp and version control.

Skills: agricultural fieldwork, agricultural practices, and teamwork.
From: February 2022. **To:** June 2022.

Employer: Plant & Food Research.

Address: Batchelar Road, Fitzherbert, Palmerston North 4474, Aotearoa New Zealand.

Position: Casual Technician.

Duties: Crop harvesting and seed cleaning.

Skills: post-1080 drop monitoring work, advanced bush fieldwork, and rat and possum line laying.

From: 15 July 2020. **To:** 22 July 2020.

Employer: Tauahika Ltd.

Address: 265 Ngaere Park Road, Turitea, Palmerston North, RD2, 4474, Manawatū-Whanganui, Aotearoa New Zealand.

Position: Fieldworker.

Duties: Post-1080 drop monitoring.

Skills: strict deadline work, teamwork, and client-oriented work.

From: 1 August 2019. **To:** 30 November 2020.

Employer: Michael and Paul Weber.

Address: Clearwater, Florida, United States of America.

Position: Translator, English to Italian.

Duties: Translating patents.

Skills: kiwi catching, handling, measuring, transmitterising, and egg candling.

Date: March 2019.

Trained by: Isabel Castro for kiwi catching, handling, measuring, and transmitterising; Emma Bean for egg candling.

Kiwi Accredited Handler number: 0403.

Date: 1 December 2020.

Course: Operation Nest Egg Candling Course.

Institution: National Kiwi Hatchery, Rotorua, Bay of Plenty, Aotearoa New Zealand.

Skills: Bayesian inference, L^AT_EX, science communication, and English language proficiency.

From: 26 July 2014. **To:** 31 July 2015.

Institution: Utah State University.

From: 1 March 2012. **To:** 30 June 2014.

Role: Italian Red Cross volunteer, emergency ambulance driver and first responder.

Institute: Italian Red Cross, Porto Potenza Picena Section, Potenza Picena, Macerata, Italy.

Skills: C programming, descriptive statistics, advanced office suite use, high- and low-level computer programming, Unix, and Linux.

From: 15 September 2001. **To:** 15 June 2004.

Institution: ITIS Montani, Fermo, Italy.

LINGUISTIC PROFICIENCIES

Italian, mother tongue.

English, TOEFL 109/120 in 2018.

Spanish, advanced.

French, rudimentary.

Papiamentu, rudimentary.

Te reo Māori, beginner.